An Oblivious Lower Bound for Online Weighted Flow Time

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Abstract

We give a proposed proof that no randomized online algorithm is constant competitive for weighted flow time on a single unit-speed machine with preemption and known processing times. The proof alternates a geometric-tail amplification with averaging, then converts unfinished weight into flow time. It constructs a finite input distribution independent of the algorithm. The proposed result awaits independent mathematical review.

1 The goal and the proof idea

Job j has release time $r_j \geq 0$, size $p_j > 0$, and weight $w_j > 0$. The algorithm learns its size and weight on arrival. It may preempt freely and may idle. Both it and the offline comparator have one unit-speed machine. For an instance I, let

$$F_A(I) = \sum_j w_j(C_j - r_j) = \int_0^\infty U_A(t)\,dt,$$

where $U_A(t)$ is the weight of released, unfinished jobs. Let $\mathrm{OPT}(I)$ denote the minimum weighted flow time. A policy that never finishes some job has infinite cost.

Theorem 1. *For every $R \geq 1$, there is a finite-support distribution $\mathcal{D}_R$ over finite instances such that every deterministic online policy A satisfies*

$$\mathbb{E}_{I\sim\mathcal{D}_R} F_A(I) > R\,\mathbb{E}_{I\sim\mathcal{D}_R}\,\mathrm{OPT}(I).$$

Consequently, no randomized online algorithm is constant competitive against an oblivious adversary. The conclusion also holds with a fixed additive constant.

Bansal and Chan[1] established a nonconstant deterministic lower bound by amplifying unfinished weight. We use the same viewpoint and construct a single distribution that works for every deterministic policy.

Proof idea. First create a gap at one time: offline leaves expected unfinished weight at most one, while every online policy leaves expected weight at least c in jobs that still need at least one unit of processing. We build this gap from independent copies, making earlier copies large enough that their unfinished jobs survive the later ones.

Two operations make c arbitrarily large. *Amplification* adds one heavy job and chooses the terminal time using fair coins. Its weight pays for the old backlog that the policy can remove, while offline gains a small advantage at one terminal. *Averaging* keeps the resulting random weight under control, so amplification can be repeated. Finally, a long stream of small jobs turns the gap at one time into a gap in total flow time.

[1] Nikhil Bansal and Ho-Leung Chan. *Weighted Flow Time Does Not Admit $O(1)$-Competitive Algorithms.* SODA, 2009, pp. 1238–1244. Author version.

2 Accumulating unfinished jobs

A *terminal instance* is a pair (I, T), where $T > 0$ and every job of I arrives strictly before T. The pair is sampled in advance; the policy sees arrivals but is not told T. The terminal may be correlated with I, and distinct outcomes may have the same arrivals but different terminals. Write

$$U^*(I, T) = \min_S U_S(T)$$

for the minimum unfinished weight at T over feasible offline prefix schedules. This minimum is attained, since only finitely many unfinished subsets are possible. Each outcome may use its own offline schedule.

Let $p^{\mathrm{rem}}_{A,j}(t)$ be job j's remaining processing under A. Our only online potential is

$$\Phi_M(A; I, T) = \min\left\{ M, \sum_{j \in I:\, p^{\mathrm{rem}}_{A,j}(T) \geq 1} w_j \right\}. \tag{1}$$

The unit residual ensures that counted jobs cannot disappear instantly. The cap M bounds how much weight one copy can contribute to the analysis; it does not change any actual job weight.

Definition 2. *For $0 < c \leq M$, a finite-support terminal distribution is (c, M)-*hard *if*

$$\mathbb{E}U^*(I, T) \leq 1, \qquad \mathbb{E}\Phi_M(A; I, T) \geq c \quad \textit{for every deterministic causal policy } A.$$

Policies may idle and have arbitrary fixed initial state, independent of the sampled outcome.

The distribution is chosen before the policy. Its full law may be known to the policy; only the realized future is unavailable. Increasing M preserves hardness. A common rescaling of weights is handled by simulation: feed the rescaled weights to the policy, and implement the same service decisions on the original jobs. Thus weight bounds scale by the same factor without assuming that the policy itself is scale invariant.

We concatenate independent copies at decreasing time scales, making the counted jobs too large for later copies to finish them. The construction serves both averaging and amplification. Write $w(E) = \sum_{j \in E} w_j$.

Lemma 3 (Separated copies)**.** *Let $\mathcal{D}$ be (c, M)-hard, and fix an integer $m \geq 1$. Choose $P \geq 1$ bounding both T and $\sum_{j \in I} p_j$ on its finite support. Choose the scales backward, for $i = m, m-1, \ldots, 1$:*

$$S_i = P \sum_{j > i} s_j, \qquad s_i = 2S_i + 2, \qquad V = P \sum_{j=1}^{m} s_j. \tag{2}$$

Thus $s_m = 2$, and S_i bounds both the duration and the total processing volume of all copies after i. Draw m independent copies (I_i, T_i). With

$$u_0 = 0, \qquad u_i = \sum_{j \leq i} s_j T_j, \qquad t_0 = u_m,$$

release copy i's jobs at $u_{i-1} + s_i r_j$, with sizes $s_i p_j$ and unchanged weights.

For any deterministic policy, let

$$E_i = \{ j \in I_i : p^{\mathrm{rem}}_{A,j}(u_i) \geq s_i \}, \qquad X_i = \min\{M, w(E_i)\}.$$

The execution may have further arrivals from t_0 onward. The following hold:

1. *If $\mathcal{F}_i$ reveals the first i complete copy draws, then X_i is $\mathcal{F}_i$-measurable and*

$$0 \le X_i \le M, \qquad \mathbb{E}[X_i \mid \mathcal{F}_{i-1}] \ge c. \tag{3}$$

2. *Every job in E_i has residual at least $S_i + 2$ at t_0.*

3. *An offline schedule leaves unfinished weight $\sum_i B_i$ among the copy jobs at t_0, where $B_i = U^*(I_i, T_i)$ and $\mathbb{E}B_i \le 1$. The total original volume of copies $j > i$ is at most S_i. All old volume and t_0 are at most V.*

Proof. For the first assertion, fix the preceding copy draws. The policy's state just before the current copy begins, its absolute start time, and all earlier jobs and residuals are now fixed. Simulate this policy on the current copy, translating the clock and job sizes by s_i and carrying earlier jobs as virtual state. Work on a virtual old job becomes idling on the current copy. Suppress every subsequent arrival *at every time.*

This defines a causal policy without knowing the current T_i. For each current outcome, the simulated and real histories agree strictly before u_i: no subsequent copy or additional job has arrived yet. Arrivals at u_i cannot change service already performed. Hence terminal residuals agree. The current draw is independent of the fixed past, so its hardness guarantee gives (3). Those residuals depend only on the first i draws, proving measurability.

The simulator never needs to detect T_i. In particular, it exhibits the same absence of arrivals at every earlier possible terminal. Its behavior after the actual terminal is irrelevant, and it may be extended arbitrarily on histories outside the support.

Since $t_0 - u_i \le S_i$, each job in E_i retains residual at least

$$s_i - (t_0 - u_i) \ge s_i - S_i = S_i + 2. \tag{4}$$

For the offline assertion, run each copy's optimal terminal schedule in its own interval, ignoring unfinished jobs from earlier copies. The stated volume and duration bounds follow from the choice of P. □

3 Averaging and amplification

Averaging reduces the cap with an arbitrarily small loss in the guarantee. The copies are sampled independently, but the policy can react to the past; the conditional guarantee in Lemma 3 is what we need.

Lemma 4 (Averaging)**.** *From a (c, M)-hard distribution with $c > 0$, for every $0 < \varepsilon < c$, one can construct a $(c - \varepsilon, c)$-hard distribution.*

Proof. Use $m = \lceil (M/\varepsilon)^2 \rceil$ copies from Lemma 3, take terminal t_0, and divide all job weights by m. Fix any policy; use the weight-rescaling simulation to analyze it before this division. Every job in E_i still has residual at least one at t_0, so with $S = \sum_i X_i$ the new potential is at least

$$\frac{1}{m} \min\{mc, S\}.$$

The variables X_i need not be independent, because the policy uses its past. Put $\mu_i = \mathbb{E}[X_i \mid \mathcal{F}_{i-1}]$ and $Z = \sum_i (X_i - \mu_i)$. The summands are bounded martingale differences. For $i < j$ their product has expectation zero by conditioning on $\mathcal{F}_{j-1}$; consequently

$$\mathbb{E}Z^2 \le mM^2, \qquad S \ge mc + Z.$$

Cauchy–Schwarz now gives

$$\begin{aligned}\mathbb{E}\min\{mc, S\} &= mc - \mathbb{E}(mc - S)_+ \\ &\geq mc - \mathbb{E}|Z| \geq mc - M\sqrt{m}.\end{aligned}$$

After division by m, the expected potential is at least $c - M/\sqrt{m} \geq c - \varepsilon$, with cap c.

The offline schedule from Lemma 3 leaves weight $m^{-1}\sum_i B_i$ after the division, with expectation at most one. The construction has finite support, and its residual threshold remains one. □

The amplifier adds a job of weight $2M$. A geometric choice of terminal leaves at most two copies in the relevant tail on average, so this one job pays for the entire tail. Offline knows the terminal and can do slightly better.

Lemma 5 (Amplification)**.** *For an integer $M \geq 1$, put*

$$k = 2M + 1, \qquad \eta = \frac{1}{2^k - 2}.$$

From any (c, M)-hard distribution one can construct a $((1+\eta)c, M')$-hard distribution for some finite M'.

Proof. Use k copies from Lemma 3. At their final time t_0 release one new job J of size V and weight $2M = k - 1$. There are no further arrivals in this terminal instance. Define

$$h_i = t_0 + V + S_i, \qquad i = 1, \ldots, k.$$

The checkpoints satisfy $h_1 > \cdots > h_k = t_0 + V$: a larger index means an earlier time. Independently of the old draws, toss $k - 1$ fair coins in advance. Let K be the index of the first head, or k if all are tails, and take terminal $T = h_K$. The choice of K changes no job or arrival.

Since $\mathbb{P}(K \geq i) = 2^{1-i}$ for $1 \leq i \leq k$, every tail satisfies

$$\mathbb{E}(K - \ell + 1)_+ = \sum_{i=\ell}^{k} \mathbb{P}(K \geq i) \leq 2\mathbb{P}(K \geq \ell) \qquad (1 \leq \ell \leq k+1). \tag{5}$$

Here an empty sum is zero. This is the statement that, conditional on reaching any index, at most two indices remain in expectation, including that index.

Online weight. Fix a policy and the old copy draws. Its trajectory with no further arrivals is the same for every K. Let Y_i be the total weight of jobs with residual at least one at h_i, and let

$$\ell = \min\{i : p^{\mathrm{rem}}_{A,J}(h_i) \geq 1\},$$

with $\ell = k + 1$ if the set is empty. As residuals only decrease with time, J contributes its weight $2M$ to Y_i exactly when $i \geq \ell$.

For $j < \ell$, every job in E_j still has residual at least one at h_j. Indeed, J has residual below one there. If a job in E_j also had residual below one, the two jobs would have received more than

$$(V - 1) + (S_j + 2 - 1) = V + S_j = h_j - t_0$$

units of service after t_0, which is impossible. These jobs also have residual at least one at every earlier checkpoint h_i with $i \geq j$.

Thus the weights X_j for $j \le K$ and $j < \ell$ are all present at h_K. The omitted tail has at most $(K - \ell + 1)_+$ terms, each at most M, whereas J contributes $2M$ whenever this tail is nonempty. Averaging over K only,

$$\begin{aligned}\mathbb{E}_K Y_K &\ge \mathbb{E}_K \sum_{j \le K} X_j - M\mathbb{E}_K(K - \ell + 1)_+ + 2M\mathbb{P}(K \ge \ell) \\ &\ge \sum_{j=1}^{k} \mathbb{P}(K \ge j) X_j,\end{aligned}$$

where the last step is (5). Now average over the old draws and use $\mathbb{E}X_j \ge c$:

$$\mathbb{E}Y_K \ge c \sum_{j=1}^{k} \mathbb{P}(K \ge j) = c\mathbb{E}K. \tag{6}$$

Offline weight. Before t_0, use the copy schedules from Lemma 3. For terminal h_i with $i < k$, complete J and then all remaining jobs from copies $j > i$. This takes at most $V + S_i$ time and leaves weight at most $\sum_{j \le i} B_j$, with expectation at most i. For terminal h_k, instead complete all old jobs in at most V time and leave J, of weight $k - 1$. Thus offline saves one unit against the benchmark K when $K = k$. Writing $\delta = \mathbb{P}(K = k) = 2^{1-k}$, we have $\mathbb{E}K = 2 - \delta$, so

$$\mathbb{E}U^*(I, T) \le \beta := \mathbb{E}K - \delta = 2 - 2\delta. \tag{7}$$

Divide all weights by the fixed positive number β. The offline expectation becomes at most one, while (6) becomes

$$\frac{c\mathbb{E}K}{\beta} = c\frac{2^k - 1}{2^k - 2} = (1 + \eta)c.$$

Choose M' to bound the total weight of every normalized input. Such a finite bound exists because the support is finite, and the cap does not truncate the potential. The weight-rescaling simulation gives the guarantee for every policy. All jobs arrive strictly before T, so the new law is hard as claimed. □

For example, $M = c = 1$ gives $k = 3$: the online bound is $7/4$ and the offline bound is $3/2$, giving a normalized guarantee of $7/6$. The cap may increase; averaging is what makes the improvement repeatable.

4 Completing the proof

Arbitrarily large terminal gaps. Fix $R \ge 1$ and an integer $M > 2R$. Keep the factor η from Lemma 5 fixed throughout. A single job of size two and weight one, released at zero with terminal one, is $(1, M)$-hard.

Suppose a (c, M)-hard law has been constructed with $1 \le c \le R$. Amplification gives guarantee $(1 + \eta)c$. Apply averaging with loss $\varepsilon = \eta c/2$. The resulting guarantee and cap are respectively

$$\left(1 + \frac{\eta}{2}\right) c \qquad \text{and} \qquad (1 + \eta)c < 2R < M.$$

Increase the cap to M. Thus a round multiplies the guarantee by the fixed factor $1 + \eta/2 > 1$, and another round is possible until it exceeds R. Finitely many rounds therefore give a (c, M)-hard law with $c > R$. Every round uses finitely many copies at fixed positive scales; the final law still has finite support and bounded job count, time, volume, and weight.

Turning the terminal gap into flow time. Fill the machine with a stream arriving at unit load. Working on old jobs creates an equal amount of stream backlog. A high weight-to-size ratio makes that backlog expensive, while small job weights keep the offline stream cost small.

Let P bound both the total processing volume and the terminal time of the final law, and let Q bound its total weight. Choose

$$\varepsilon = \frac{c-R}{2R} > 0, \qquad d = \frac{\varepsilon}{M} > 0.$$

For a fixed integer n to be chosen, set $L = nd$. At the sampled terminal T, append n jobs of size d and weight ε, released at $T, T+d, \ldots, T+(n-1)d$. All parameters, including n, are independent of the realized outcome and the policy.

Fix any deterministic policy on these extended inputs. Suppressing every stream arrival indefinitely gives a causal policy on the old law, with the same old-job residuals at T, by the simulation in Lemma 3. Hence, for the old jobs,

$$H := \Phi_M(A; I, T) \le M, \qquad \mathbb{E}H \ge c.$$

At time $T+s$, $0 \le s < L$, let $x(s)$ be service given to old jobs since T. The stream has released volume $d(\lfloor s/d \rfloor + 1) \ge s$, and received service at most $s - x(s)$. Its residual volume is at least $x(s)$, so its unfinished weight is at least

$$\frac{\varepsilon}{d} x(s) = Mx(s).$$

If $x(s) < 1$, every old job with residual at least one at T remains unfinished, so old unfinished weight is at least H. If $x(s) \ge 1$, stream unfinished weight is at least $M \ge H$. Thus total unfinished weight is at least H throughout the stream, giving

$$\mathbb{E}F_A \ge L\mathbb{E}H \ge cL. \tag{8}$$

Offline follows a prefix attaining $U^*(I,T)$, processes each stream job immediately upon release, and then clears the old jobs. Old jobs contribute at most QP before the stream, $U^*(I,T)L$ during it, and QP during cleanup. Stream jobs contribute exactly εL. Consequently

$$\mathbb{E}\,\mathrm{OPT} \le 2PQ + (1+\varepsilon)L. \tag{9}$$

Combining (8) and (9),

$$\mathbb{E}F_A - R\mathbb{E}\,\mathrm{OPT} \ge \big(c - R(1+\varepsilon)\big)L - 2RPQ = \frac{c-R}{2}L - 2RPQ.$$

Since $c > R$, a sufficiently large fixed n makes this positive, uniformly over all deterministic policies. This proves the first claim of Theorem 1.

Randomized algorithms. Fixing a randomized algorithm's private random tape gives a deterministic policy. The input law is independent of that tape, so averaging the same uniform gap gives

$$\mathbb{E}_I\mathbb{E}_{\mathrm{coins}}F_A(I) > R\mathbb{E}_I\,\mathrm{OPT}(I).$$

The finite support must therefore contain an input I with $\mathbb{E}_{\mathrm{coins}}F_A(I) > R\,\mathrm{OPT}(I)$. If a fixed additive constant Γ is permitted, choose n so that the uniform gap exceeds Γ; the same averaging argument defeats it. This proves the theorem. □

5 Dependence on the size and weight ratios

Write

$$P(I) = \frac{\max_j p_j}{\min_j p_j}, \qquad W(I) = \frac{\max_j w_j}{\min_j w_j}.$$

Here $P(I)$ is a ratio; below we use H for the volume and time bound called P in the preceding construction. All logarithms in this section are base two.

Theorem 6. *There is an absolute constant C such that, for every integer $r \geq 2$, there is a finite-support input distribution satisfying*

$$\mathbb{E}F_A > r\,\mathbb{E}\,\mathrm{OPT} \quad \textit{for every deterministic online policy } A,$$

whose every input obeys

$$W(I) \leq 2^{2^{Cr}}, \qquad P(I) \leq 2^{2^{2^{Cr}}}. \tag{10}$$

Consequently, for sufficiently large bounds P, W, the randomized competitive ratio on instances with $P(I) \leq P$ and $W(I) \leq W$ is

$$\Omega(\min\{\log\log W,\ \log\log\log P\}).$$

Proof. A cap independent of the total weight. The only refinement we need is that the output cap in Lemma 5 can be chosen as

$$B = (k+1)M = 2M(M+1).$$

To see this, fix the old draws and use the threshold ℓ from that proof. The contribution guaranteed at checkpoint i is

$$Z_i = \sum_{j \leq \min\{i, \ell-1\}} X_j + 2M\,\mathbf{1}_{\{i \geq \ell\}}.$$

If $i < \ell$, at most k copy contributions are counted. Otherwise at most $k-1$ are counted, together with J. Thus $Z_i \leq B$ in both cases. Since $Y_i \geq Z_i$, the online argument remains valid with Y_i replaced by $\min\{B, Y_i\}$. Also $1 < \beta < 2$, so after normalization,

$$\min\{B, Y_i/\beta\} \geq \frac{1}{\beta}\min\{B, Y_i\}.$$

This proves the claimed cap bound without any dependence on the total weight.

A fixed number of copies per round. Set

$$M = 4r, \qquad k = 2M+1, \qquad \eta = \frac{1}{2^k - 2}, \qquad m = \left\lceil \left(\frac{2B}{\eta}\right)^2 \right\rceil.$$

Start from the single-job law with guarantee $c_0 = 1$. In each round amplify, then average m copies with loss $\eta/2$. For $c_s \geq 1$, the resulting guarantee is at least

$$(1+\eta)c_s - \eta/2 \geq (1+\eta/2)c_s.$$

Thus we may take $c_s = (1+\eta/2)^s$. As long as $c_s < 2r$, the cap after averaging is $(1+\eta)c_s < 4r = M$, and may be increased to M. Stop after the first t rounds with $c_t \geq 2r$. Then

$$t = \left\lceil \frac{\ln(2r)}{\ln(1+\eta/2)} \right\rceil = 2^{O(r)}, \qquad \log m = O(r), \qquad k = O(r). \tag{11}$$

Indeed, $\eta^{-1} = 2^{O(r)}$, $B = O(r^2)$, and $\ln(1 + \eta/2) \geq \eta/4$.

Weights. Every round introduces jobs of weight $2M$ before dividing all weights by the same factor βm. Here β is fixed across rounds and lies in $(1, 2)$. Starting from weight one, induction gives, after t rounds,

$$(\beta m)^{-t} \leq w_j \leq 2M. \tag{12}$$

The averaging operation does not multiply the total weight bound by m, because it also divides every weight by m.

Processing times. Let $H_s \geq 2$ bound the total processing volume and terminal time after s rounds, with $H_0 = 2$. The backward scaling recursion for a copies gives total volume and duration at most

$$V = (2H_s + 1)^a - 1.$$

Amplification therefore permits the intermediate bound $\widehat{H}_s = 3((2H_s + 1)^k - 1)$: all old jobs and J have total volume at most $2V$, and every terminal is at most $3V$. The following averaging step permits

$$H_{s+1} = (2\widehat{H}_s + 1)^m - 1 \leq \left(9(3H_s)^k\right)^m \leq (9H_s)^{km},$$

where the last inequality uses $k \geq 2$. Consequently,

$$\log(9H_t) \leq (2km)^t \log 18. \tag{13}$$

Every old job has size at least one: copy scales are at least two, weight normalization does not change sizes, and every new job J has size $V \geq 1$.

The final stream. Let Q bound the total weight of the final terminal law. Append stream jobs of weight $1/2$ and size $d = 1/(2M)$, one every d time units, for a fixed duration $L = nd > 4QH_t$. The argument in Section 4 gives

$$\mathbb{E}F_A \geq 2rL, \qquad \mathbb{E}\,\mathrm{OPT} \leq 2QH_t + \tfrac{3}{2}L,$$

and hence $\mathbb{E}F_A > r\mathbb{E}\,\mathrm{OPT}$ for every deterministic policy. The length of the stream changes neither ratio. Since $t \geq 1$ and $(\beta m)^{-t} < 1/2$, (12) yields

$$W(I) \leq 2M(2m)^t, \qquad P(I) \leq 2MH_t.$$

Together with (11) and (13), these imply

$$\log W(I) \leq 2^{O(r)}, \qquad \log\log P(I) \leq 2^{O(r)}.$$

Increasing a fixed absolute C gives (10). Choosing $r = \Theta(\min\{\log\log W, \log\log\log P\})$ small enough to respect both parameter bounds, and averaging over the algorithm's random tape, proves the competitive-ratio claim. An additive constant is handled by lengthening the stream, without changing either ratio. □

For comparison, Bansal–Chan's deterministic lower bound is

$$\Omega\left(\min\left\{\sqrt{\frac{\log W}{\log\log W}}, \quad \sqrt{\frac{\log\log P}{\log\log\log P}}\right\}\right).$$

The weaker dependence here comes from the amplification gain $\eta = 2^{-\Theta(r)}$, which requires exponentially many rounds. The parameter count above quantifies the randomized construction; it does not recover the deterministic rate.